

EXERCISE 1 - SQL FOUNDATIONS

01 Selecting & Sorting

01. SELECT *
FROM employees;

Table employees

id	first_name	last_name	department	Salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02. SELECT DISTINCT department
FROM employees;

Table employees

department
IT
HR
Finance
Marketing

03. SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

TABLE employees

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04. SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

Employees

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

02. Filtering with WHERE

05. SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

employees

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06. SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;

employees

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

07. SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';

employees

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08. SELECT id, first_name, last_name, department
FROM employees
WHERE department <> 'IT';

employees

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09. SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

employees

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';

employees

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg